

Ticks, Volume, and Dollar Bars: Are They Truly "Information-Driven"?

By Ahmed Nabil — Lead AI Engineer @ Prometheus-IQ | AI Researcher | Quant Finance enthusiast · LinkedIn post

Ticks, volume, and dollar bars are mathematically superior to arbitrary time-bars [1]. But are they truly "information-driven"?

Not necessarily.

In this series, we don't guess—we validate with data. Let's look at our 3-month SIP trade dataset [2]. In our first figure, you can clearly see:

-  A heavily skewed size distribution: Our size distribution is heavily left-skewed. On average, 70.8% are odd-lot prints. These are highly noisy transactions with no aggressive trading sign.
-  Massive right-tail block trades: We also see massive right-heavy tails consisting of institutional block trades exceeding 1,000,000 shares.

Standard volume-bar sampling fails here. Look at the severe mismatch between share rank and dollar rank across our portfolio:

- AAPL: share_rank=2 (median size=14.0) vs. dollar_rank=5 (median \$=4088.1)
- ASML: share_rank=5 (median size=4.0) vs. dollar_rank=1 (median \$=6583.6)
- GOOG: share_rank=1 (median size=15.0) vs. dollar_rank=3 (median \$=5346.6)
- AMAT: share_rank=3 (median size=10.0) vs. dollar_rank=2 (median \$=5757.5)
- TSM: share_rank=3 (median size=10.0) vs. dollar_rank=4 (median \$=4329.0)

Can standard tick sampling fix this?

Data says no. Look at the median dynamics:

- ASML: median size = 4 shares | median dollar ≈ \$6,735
- AAPL: median size = 13 shares | median dollar ≈ \$3,859

An AAPL trade carries far less economic value than ASML, yet tick sampling treats both as exactly "1 tick." It measures pure transaction frequency, making it a poor proxy for true economic or information flow on an internalized, odd-lot-heavy tape.

So, are standard dollar bars the answer?

Yes and no. They capture transacted value well; speeding up when the tape is heavy and slowing down when it is quiet—but they fail to isolate asymmetric information flow. Dollar-Runs bar doesn't close on total transacted value alone. It closes when the directional persistence of order flow becomes statistically surprising.

Comparing Dollar-Runs to standard dollar bars (shown in the plots), three critical elements stand out:

- ♦ **Dynamic Frequency:** Bars generate only when signed print expectations are exceeded, capturing shifting information density across the portfolio.
- ♦ **Order Flow Direction:** Unlike standard dollar bars that blindly pool all volume, runs isolate the true asymmetry between buy and sell sequences.
- ♦ **Event Detection:** Runs capture structural order book changes before a standard dollar bar breaches its threshold, visible via the theta-to-expectation ratio in the third chart.

I noticed some great questions in the last post regarding alternative setups like Renko bars. I have answered those directly in the comments below.

Disclaimer: For educational purposes only. Not investment advice.

Source: https://www.linkedin.com/posts/ai-ahmed_ticks-volume-and-dollar-bars-are-mathematically-ugcPost-7485089005194166273-UEF-/